Hoang Anh Nguyen Huu

Robotics / Control Engineering Intern

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SUMMARY

Control Engineering and Automation undergraduate with hands-on experience in ROS 2, mobile robotics, embedded systems, and robot control. Seeking a robotics internship focused on robot software, perception, and motion control.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT) — Hanoi, Vietnam

B.Eng. Candidate in Control Engineering and Automation — Expected Graduation: 2028

- Specialization: **Robotics and Applied AI**; Current GPA: **3.47/4.0**
- **1st Prize**, PTIT Robot Programming Competition (2025)
- **Sakura Science Exchange Program** – JST Scholarship, PTIT × Gunma University, Japan (Feb 2025)
- **Encouragement Award**, EC Challenge Robot Programming Competition(2024)

TECHNICAL SKILLS

Robotics: ROS 2, Gazebo, RViz, Nav2, SLAM, Forward/Inverse Kinematics

Programming: C/C++, Python, MATLAB/Simulink

Perception: OpenCV, YOLO

Embedded & Tools: STM32, ESP32, FreeRTOS, Linux, Git, CMake

EXPERIENCE & PROJECTS

DTT Technology Joint Stock Company (DTT) — Apr 2026 – Present

R&D Engineer (promoted from R&D Intern) — Hanoi, Vietnam

- Developed an end-to-end YOLOv8n pipeline for Leanbot detection and 360-degree orientation estimation, including automated data collection and labeling, 24-class training, and custom Soft Angular BCE with circular vector-based angle decoding.
- Optimized real-time CPU inference using OpenVINO and ROI tracking; benchmarked PyTorch, ONNX Runtime, and OpenVINO FP32/FP16/INT8 variants, improving throughput from 8.4 FPS to approximately 31 FPS and reducing inference latency from approximately 21 ms to 6 ms.

Vision-Guided Navigation for TurtleBot 4 Lite | *ROS 2, Nav2, SLAM, YOLOv8, OAK-D, Gazebo*

- Developed a modular ROS 2 pipeline integrating RGB-D object detection, 3D object localization, autonomous patrol, and Nav2-based goal generation.

Micromouse Maze-Solving Robot | *STM32F411, PID Control, Flood Fill, A**

- Developed bare-metal firmware integrating encoder, IMU, and IR sensing with PID motion control, maze exploration, optimal-path planning, flash storage, and BLE debugging.

Quadcopter Flight Controller | *ESP32, FreeRTOS, PID Control, Complementary Filter, BLE*

- Developed modular flight-controller firmware with MPU6050-based attitude estimation using complementary filtering, PID roll/pitch/yaw stabilization, PWM motor control, and BLE configuration.

6-DOF FAIRINO Robot Modeling and Motion Control | *Kinematics, Dynamics, Trajectory Planning*

- Developed forward/inverse kinematics, Jacobian analysis, and joint-space and Cartesian trajectory planning for a 6-DOF FAIRINO industrial robot.
- Integrated trajectory commands with the FAIRINO SDK and validated end-effector motion on the physical robot; evaluated PD control with dynamic compensation in MATLAB/Simulink.

LANGUAGES

English: CEFR B2 (Aptis ESOL) | **Japanese**: JLPT N3